

packages/graph-core/src/staleness.ts

Kind: added · Baseline: a9c34bfb03fe · Target: NovaDiff · Generated: 2026-05-20T05:28:00.890Z

Overview

A new module `packages/graph-core/src/staleness.ts` (lines 1-90) adds utilities for detecting stale knowledge graphs and merging incremental updates.

Key changes

- **Imports** (R1-R2): `execFileSync` from `child_process` and type definitions from `./types.js`.
- **StalenessResult interface** (R4-R7): exposes `stale: boolean` and `changedFiles: string[]`.
- **getChangedFiles** (R13-R28): runs `git diff <lastCommitHash>..HEAD --name-only` in `projectDir`; returns trimmed file paths or an empty array on error.
- **isStale** (R34-R42): calls `getChangedFiles` and reports `stale` if any files changed.
- **mergeGraphUpdate** (R54-R90):
 1. Builds a set of changed file paths (`changedSet`).
 2. Removes nodes whose `filePath` matches a changed file (`removedNodeIds`).
 3. Keeps edges whose source and target are not in `removedNodeIds`.
 4. Returns a new graph with updated `project.gitCommitHash` and `project.analyzedAt`, and appends `newNodes` / `newEdges`.

Impact

- **Correctness:** Provides deterministic staleness checks and a clear merge strategy for incremental graph updates.
- **Maintainability:** Centralizes git-based change detection; future extensions can reuse `getChangedFiles`.
- **Performance:** `execFileSync` blocks the event loop; suitable for CLI tooling but may need async refactor for high-throughput services.
- **Compatibility:** Relies on a Git repository; non-Git environments will silently return an empty change list.

Risks & follow-ups

- **Error handling:** `getChangedFiles` swallows all errors; verify that silent failures are acceptable or add logging.
- **Node removal logic:** Assumes `filePath` is defined on nodes; test with nodes lacking this property to ensure no unintended removals.
- **Edge filtering:** Only checks source/target IDs; confirm that edges referencing removed nodes are correctly pruned.

- **Timestamp precision:** `new Date().toISOString()` may not match existing format; run integration tests to confirm consistency.